

Bishop's University

CS 596 – Special Topics on Deep Learning

Final Project: Exploring Recurrent Neural Networks (RNNs) with PyTorch

Group : FP B

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Here is a brief summary description of our implemented RNN (Recurrent Neural Network) network:

Model Architecture:

The model is implemented using the StockLSTM class, which inherits from PyTorch's nn.Module class.

The model uses an LSTM (Long Short-Term Memory) layer as the core of the RNN architecture.

Input and Output Dimensions:

The input_size parameter is set to 1, indicating that the input to the LSTM layer has a single feature dimension.

The output_size parameter is also set to 1, suggesting that the model is designed to output a single value, which in this case is the predicted stock price.

Hidden Layer Size:

The hidden_layer_size parameter determines the size of the hidden state and cell state in the LSTM layer.

We explore different hidden layer sizes, such as 50, 75, and 100, as part of the hyperparameter tuning process.

Initialization of Hidden and Cell States:

The hidden_cell attribute is initialized with zero tensors of size (1, 1, hidden_layer_size) for both the hidden state and cell state.

This initialization is performed at the start of each forward pass of the model.

Forward Pass:

In the forward method, the input sequence is passed through the LSTM layer, and the output is then passed through a linear layer to produce the final prediction.

The LSTM layer's hidden and cell states are updated during the forward pass and stored in the hidden_cell attribute for the next input sequence.

Loss Function and Optimization:

We use the Mean Squared Error (MSE) loss function to calculate the loss between the predicted and actual stock prices.

The model is optimized using Stochastic Gradient Descent (SGD) with the specified learning rates (0.001, 0.01, 0.1) as part of the hyperparameter tuning process.

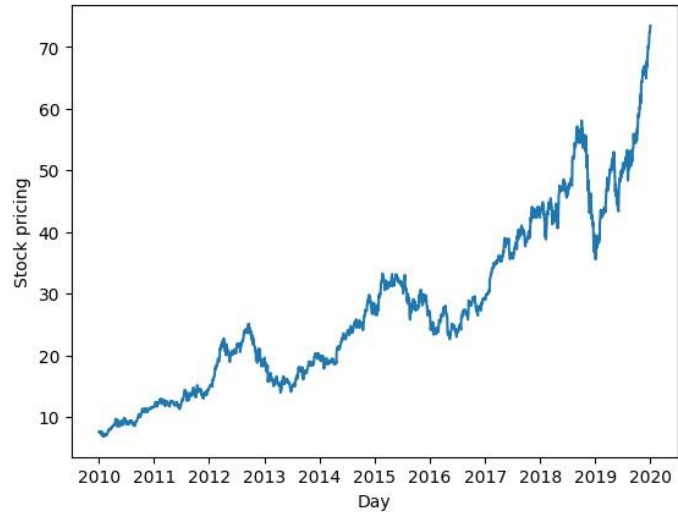
Data Preparation:

The input sequences (X) are created by sliding a fixed-size lookback window over the preprocessed stock data.

The output labels (y) are the corresponding stock prices for each input sequence.

In summary, the RNN network is an LSTM-based model designed to predict stock prices for Yahoo finance dataset. The hyperparameters, such as learning rate, hidden layer size, and batch size, are tuned to find the optimal configuration for the task.

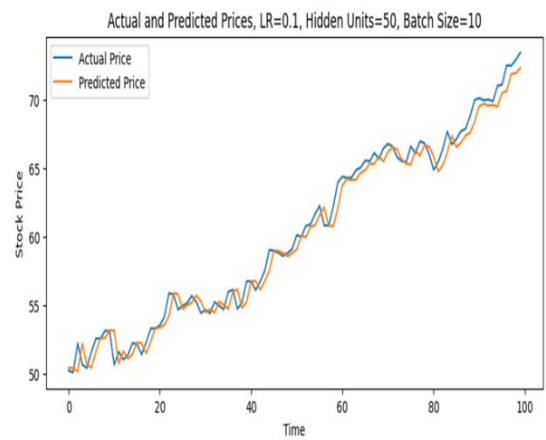
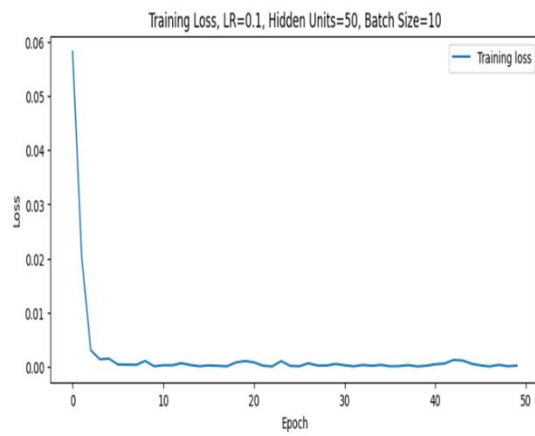
Results:



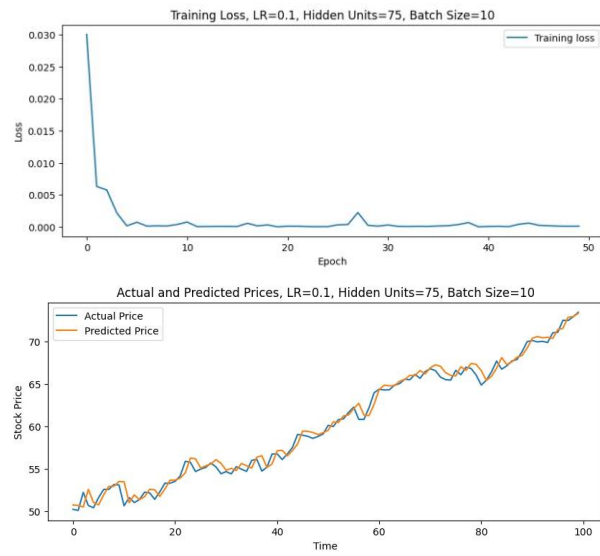
Date	Open	High	Low	Close	Adj Close	Volume \
1997-01-09	0.158482	0.159598	0.156250	0.158482	0.134171	446656000
1997-01-10	0.157366	0.162946	0.157366	0.162946	0.137950	353718400
1997-01-13	0.165179	0.165179	0.161830	0.161830	0.137005	305748800
1997-01-14	0.164063	0.164063	0.158482	0.159598	0.135116	255774400
1997-01-15	0.160714	0.160714	0.152902	0.154018	0.130392	433092800

Date	Close(t-1)	Close(t-2)	Close(t-3)	Close(t-4)	Close(t-5)
1997-01-09	0.157366	0.156250	0.159598	0.194196	0.187500
1997-01-10	0.158482	0.157366	0.156250	0.159598	0.194196
1997-01-13	0.162946	0.158482	0.157366	0.156250	0.159598
1997-01-14	0.161830	0.162946	0.158482	0.157366	0.156250
1997-01-15	0.159598	0.161830	0.162946	0.158482	0.157366

Results



Results



Results

